Workplace Development and Skills (CS3306)

Assignment: Terraform + KIND + Helm Deployment
(Due: February 27 2026. Marks: 50)

Overview

In this assignment, you will design, containerize, and deploy a personalized web application onto a Kubernetes KIND cluster created using Terraform. Your application must meet the following criteria:

- It must be a web application accessible through a browser.
- It must include your name and student ID. You may use text, images, tables, or any other suitable elements to build your web application.
- You must not include any sensitive personal information (passwords, personal email, private image, documents, etc.).

You may use AI tools to generate UI templates, styling, or layout, but all Docker, Helm, and Terraform configuration must be your own work and fully understood by you.

Prerequisites

Before starting the assignment, ensure the following tools are installed and working correctly:

Docker *KIND* *K8s* *Terraform* *Helm*

Assignment Requirements

You will build a complete deployment workflow consisting of:

1. Web Application of Your Choice

- You may develop the application in any programming language or runtime. Can be Python/Flask, Node.js, Nginx, Go, Java, etc.
- The application must run inside a Docker container.
- AI-generated UI templates or styling of web-application are permitted.
- Your application must present a web interface via HTTP (e.g., HTML page, API endpoint, etc.).

2. Docker Containerisation

- Package your application into a Docker image.
- Push the image to your own Docker Hub account.
- Ensure the image is publicly accessible.
- Include the Docker Hub URL of the image in your report.

3. Kubernetes KIND Cluster via Terraform

- The Kubernetes cluster must be created using Terraform, not via manual commands.
- The cluster name must include your name
- The cluster must include a minimum of 3 worker nodes.

4. Helm Chart

Create a Helm chart (either manually or using helm create).
Your chart must include:

- Chart.yaml

- values.yaml
- A Deployment and Service file:
  - templates/deployment.yaml
  - templates/service.yaml
- Additional template files (if needed)

Deployment requirements:

- Must pull your application image from Docker Hub that you pushed during Docker Containerisation.
- Must use more than 2 replicas (>2)

5. Port Forwarding

You must verify your deployment using port forwarding. You may use any valid port number of your choice:

kubectl port-forward svc/<service-name> 8080:80

after the above step, your application must be accessible in the browser via local host.

6. deploy.sh

A template script named **deploy.sh** is provided. The examiner will execute this script to verify your deployment. You must edit the script and insert all required details exactly as instructed in the template. Make sure that it runs end-to-end before submission. This script must be placed at the root of the submission, alongside the docker/, terraform/, and helm/ folders.

The script will:

1. Create the KIND cluster using Terraform
2. Package your Helm chart
3. Deploy your application
4. Port-forward the service

7. Required Folder Structure

You **must strictly follow** this folder structure:

```
student-id-assignment/
│
├── docker/
│   └── (Dockerfile, source code, etc.)
│
├── helm/
│   └── <your-app-name>/
│       ├── Chart.yaml
│       ├── values.yaml
│       └── templates/
│           ├── deployment.yaml
│           ├── service.yaml
│           └── other files if required
│
├── terraform/
│   └── (Terraform files for KIND cluster)
│
├── deploy.sh
│
└── README.md   (optional)
```

Submission Requirements

The mentioned submission deadline in Canvas is a hard deadline. Request for project submissions after the submission deadline won't be granted (unless there is a valid reason). You should submit your work in the Assignment section on the CS3306 area on Canvas.

You must submit **four separate files**:

1. **Source Code ZIP file:** A Zip file containing all source files that follow the above folder structure**
2. **Report Markdown ZIP File:** You are required to write the report in Markdown format. A separate ZIP file containing the main Markdown report file (e.g. student-name-report.md), any additional Markdown files (if used), and all associated assets (e.g. images, diagrams, and screenshots),
3. **PDF Report:** A PDF version generated from the Markdown source, submitted separately.
    - Filename: student-name-report.pdf
4. **A screen recording** demonstrating execution of deploy.sh

****Note:** Do not include any junk, temporary, or cached files in the submitted ZIP file, as these may cause issues during compilation and evaluation, potentially affecting your grade. Additionally, do not include .tgz file generated by the helm package command.

Report Requirements

The report (maximum 1200 words) must include:

- Introduction to the application you built
- A brief explanation of your K8s cluster
- Overall workflow
- Challenges encountered and solutions (if any)
- Personal reflection
- Required screenshots

Required Screenshots

You must include screenshots of the following:

- Docker image built locally (docker build)
- Terraform execution and KIND cluster creation
- Successful cluster creation (kubectl get nodes)
- Helm installation command and output
- Deployment output (kubectl get pods and kubectl get svc)
- The web application running in your browser

Assessment Breakdown

Component	Weight
1. Quality of the Report	30%
2. Implementation & Code Base (Docker/Terraform/Helm)	30%
3. Quality of the Web Application	10%
4. Following Instructions	10%
5. Correct Execution of deploy.sh	20%